📘 INTRODUCTION

The IPL (Indian Premier League) is an exciting tournament for cricket fans every year, generating data on team performance, player records, and match results.

The aim of this project is to analyze IPL 2022 match data using Python libraries like Pandas, NumPy, Matplotlib, and Seaborn.

Key focus areas of the project:

- Analysis of each team's win
- Relationship between the toss decision and the match result
- Ranking of top bowlers and top scorers
- Breakdown of Player of the Match awards

This analysis provides valuable insights for cricket fans, analysts, and strategists.

🎯 OBJECTIVE

- Visualize how many matches each team won in IPL 2022
- Understand the relationship between the toss decision (field or bat) and the match result
- Identify the top 10 bowlers and top 10 scorers
- Observe the distribution of Player of the Match awards
- Analyze the match winning pattern—chasing vs. defending

❓ PROBLEM STATEMENTS

1. Which team won the most matches in IPL 2022?
2. How does the fielding or batting decision after winning the toss affect the match result?
3. Who were the top 10 bowlers of IPL 2022?
4. Which player won the most "Player of the Match" awards?
5. Who were the top 10 highest scorers in IPL 2022?
6. Which strategy was more successful, chasing or defending?

🛠️ TOOLS AND TECHNOLOGY USED

Python (Pandas, NumPy, Matplotlib, Seaborn)

- Read and clean the dataset
- Exploratory Data Analysis (EDA)
- Bar charts, line charts, count plots to visualize insights
- Grouping and sorting operations for top players analysis

📌 Note: Only Python has been used in this project — MySQL and Power BI have not been used.

🔍 METHODOLOGY

Step 1: Data Collection

- IPL 2022 dataset was used in CSV format.

Step 2: Data Preprocessing

- Missing values handled
- String formatting and mapping (e.g., 'Wickets' → 'Chasing')
- Columns renamed and cleaned for clarity

Step 3: Exploratory Data Analysis

- Matches won by each team — bar chart
- Toss decision vs match result — count plot
- Top bowlers — horizontal count plot
- Player of the Match — bar chart
- Top scorers — bar chart using groupby and sorting

📈APPLICATIONS OF THE PROJECT

1. Team Performance Evaluation

- Performance can be compared by looking at the jet count of each team.

2. Toss Strategy Analysis

- By comparing the relation between toss decision and match result, better strategies can be made.

3. Player Ranking

- Player selection and training decisions can be taken by identifying top bowlers and scorers. can't

4. Fan Engagement

- Visual insights help fans understand the highlights of the tournament

🔮 FUTURE SCOPE

- *Machine Learning:* Predict which team will win based on past data
- *Interactive Dashboards:* Create real-time dashboards using Power BI or Streamlit
- *More Seasons:* Comparative analysis with data of 2021, 2023
- *Player Form Tracker:* Graphing player form from time-series analysis.

✅ EXPECTED OUTPUT

- Bar chart showing matches won by each team
- Count plot showing match wins by chasing vs defending
- Top 10 bowlers list with count
- Top 10 scorers with highest individual scores
- Player of the Match awards distribution

🏁 CONCLUSION

Analysis of IPL 2022 data revealed that:

- Gujarat won the most matches
- Both chasing and defending strategies were almost equally successful
- Bowlers like Yuzvendra Chahal and Rashid Khan were top performers
- Quinton de Kock scored the highest individual score
- Players like Jasprit Bumrah and Hardik Pandya won multiple Player of the Match awards

These insights are informative for cricket fans and can be helpful for teams in strategic planning.

In [ ]:

Out[ ]:

📌 1⃣ Import Libraries

In [1]:
```
# Importing Required Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:

Out[ ]:

📌 2⃣ Read Dataset

In [2]:
```
Dataset.read_csv("IPL 2022.csv")
data.head(5)
```

Out[2]:

	match_id	date	venue	team1	team2	stage	toss_winner	toss_decision	first_inns_score	first_inns_wkts	second_inns_score	second_inns_wkts	match_winner	won_by	margin	player_of_the_match	top_scorer	highscore	best_bo
0	1	March 26,2022	Wankhede Stadium, Mumbai	Chennai	Kolkata	Group	Kolkata	Field	131	5	133	4	Kolkata	Wickets	6	Umesh Yadav	MS Dhoni	50	Da S
1	2	March 27,2022	Brabourne Stadium, Mumbai	Delhi	Mumbai	Group	Delhi	Field	177	5	179	6	Delhi	Wickets	4	Kuldeep Yadav	Ishan Kishan	81	Kul Y
2	3	March 27,2022	Dr DY Patil Sports Academy, Mumbai	Banglore	Punjab	Group	Punjab	Field	205	2	208	5	Punjab	Wickets	5	Odean Smith	Faf du Plessis	88	Mohan
3	4	March 28,2022	Wankhede Stadium, Mumbai	Gujarat	Lucknow	Group	Gujarat	Field	158	6	161	5	Gujarat	Wickets	5	Mohammed Shami	Deepak Hooda	55	Mohan S
4	5	March 29,2022	Maharashtra Cricket Association Stadium,Pune	Hyderabad	Rajasthan	Group	Hyderabad	Field	210	6	149	7	Rajasthan	Runs	61	Sanju Samson	Aiden Markram	57	Yuzve Ch

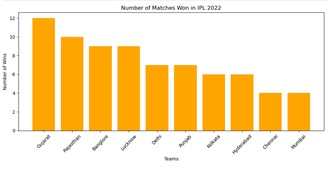
In [ ]:

Out[ ]:

📌 3⃣ Matches Won by Each Team (Matplotlib)

In [3]:
```
# Count matches won by each team
wins = Data['match_winner'].value_counts()

plt.figure(figsize=(10,5))
plt.bar(win.index, wins.values, color='orange')
plt.xticks(rotation=45)
plt.title("Number of Matches Won in IPL 2022")
plt.xlabel("Teams")
plt.ylabel("Number of Wins")
plt.tight_layout()
plt.show()
```



In [ ]:

Out[ ]:

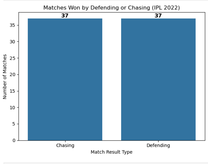
📌 4⃣ Matches Won by Defending or Chasing (Seaborn)

In [4]:
```
Data = pd.read_csv("IPL 2022.csv")
Data['won_by'] = Data['won_by'].str.strip().str.title().map({'Wickets':'Chasing','Runs':'Defending'}).fillna('Other')

ax = sns.countplot(x='won_by', data=Data, order=Data['won_by'].value_counts().index)

# Add count numbers on top of bars
for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='bottom', fontsize=12, fontweight='bold')

plt.title("Matches Won by Defending or Chasing (IPL 2022)")
plt.ylabel("Number of Matches")
plt.xlabel("Match Result Type")
plt.tight_layout()
plt.show()
```

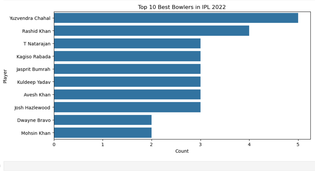


In [ ]:

Out[ ]:

🎯 5⃣ Best Bowling Players (Seaborn)

In [5]:
```
plt.figure(figsize=(10,5))
sns.countplot(
    y='best_bowling',
    palette='Blues',
    order=Data['best_bowling'].value_counts().index[:10]
)
plt.title("Top 10 Best Bowlers in IPL 2022")
plt.xlabel("Count")
plt.ylabel("Players")
plt.show()
```



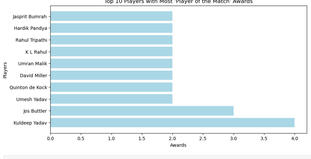
In [ ]:

Out[ ]:

🥇 6⃣ Player of the Match (Matplotlib)

In [6]:
```
# Count top 10 players with most "Player of the Match" awards
player_counts = Data['player_of_the_match'].value_counts().head(10)

plt.figure(figsize=(10,5))
plt.barh(player_counts.index, player_counts.values, color='lightblue')
plt.title("Top 10 Players with Most 'Player of the Match' Awards")
plt.xlabel("Awards")
plt.ylabel("Players")
plt.show()
```



In [ ]:

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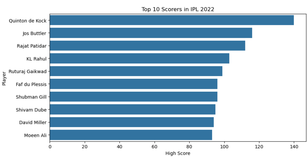
In [ ]:

💥 7⃣ Top Scorers (Seaborn + NumPy)

In [10]:
```
top_scorers = Data.groupby("top_scorer")["highscore"].max().sort_values(ascending=False).head(10)
plt.figure(figsize=(10,5))
sns.barplot(x=top_scorers.values, y=top_scorers.index)
plt.title("Top Scorers in IPL 2022")
plt.xlabel("High Score")
plt.ylabel("Player")
plt.show()
```


Top 10 Scorers in IPL 2022
Quinton de Kock
Jos Buttler
Rajat Patidar
KL Rahul
Ruturaj Gaikwad
Faf du Plessis
Shubman Gill
Shivam Dube
David Miller
Moeen Ali
Player
0
20
40
60
80
100
120
140
High Score

In [ ] :